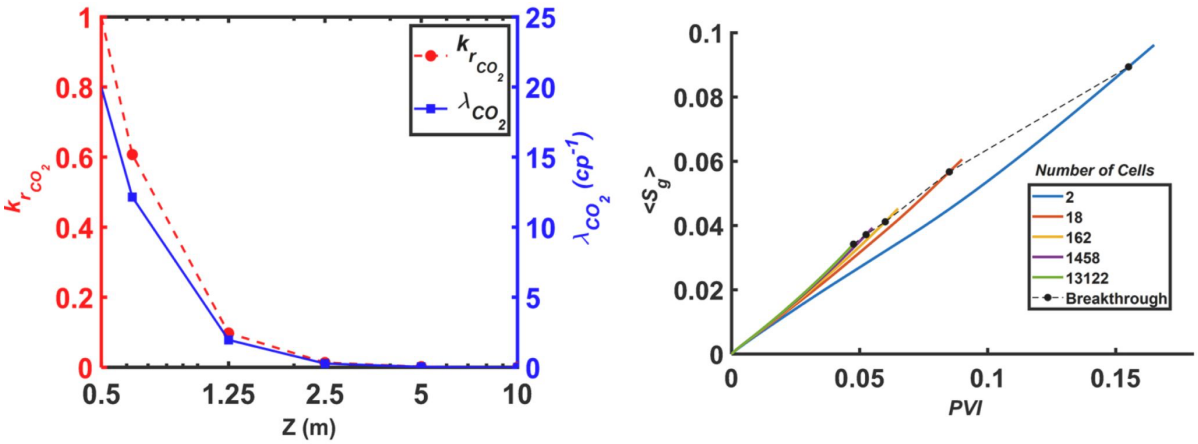
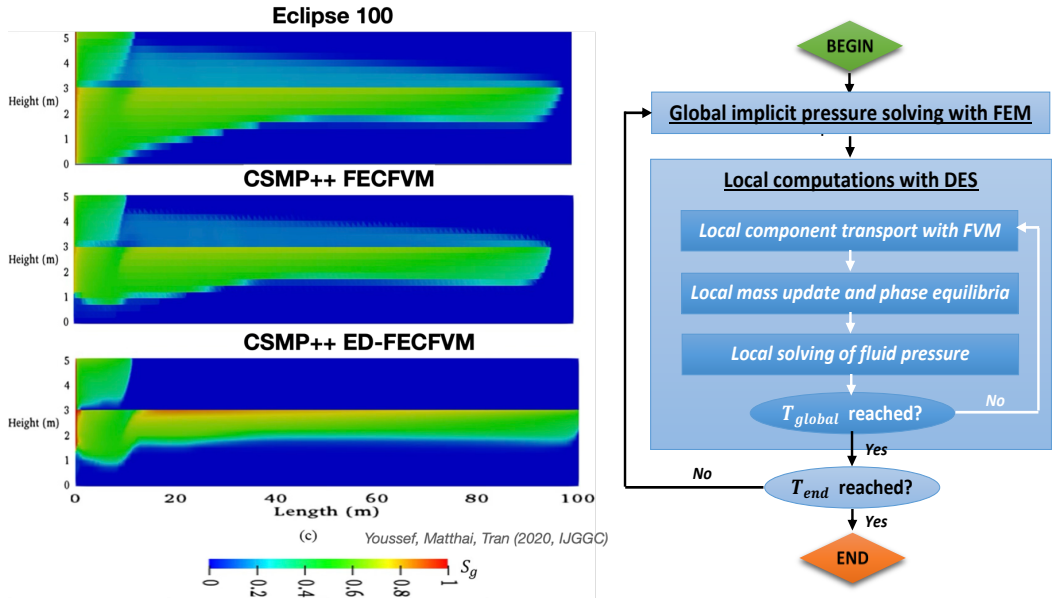
Carbon geo-sequestration

Is CO_2 plume spreading below caprocks modelled accurately? – Keeping injection plumes within lease boundaries is one of the main challenges of geosequestration. In highly permeable storage horizons, the saturation of buoyant CO_2 accumulating beneath the cap rock, should almost reach the irreducible water saturation. If so, CO_2 mobility will be high, driving fast plume spreading. In commercial reservoir simulators, however, saturation has a first order dependence on vertical grid resolution. As documented by Youssef, Matthai et al. (2021, ICGGC) this strongly affects estimates of plume breakthrough to the lease boundary and CO_2 distribution in storage horizons.



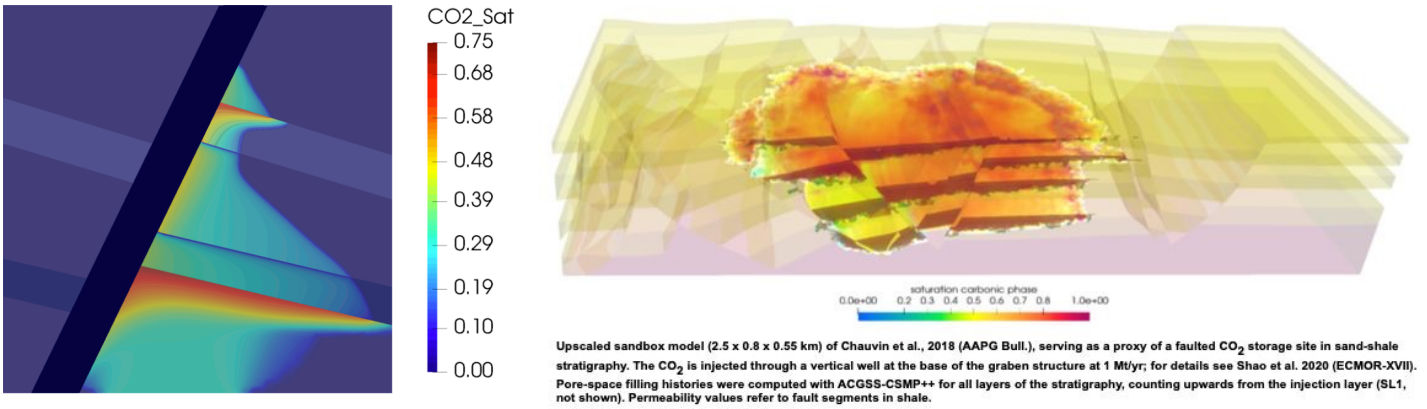
Plume arrival (right) as function of vertical grid resolution as simulated with a standard reservoir simulator that discretizes saturation as piece-wise constant in space, reducing CO_2 relative permeability (left) by volume averaging.



Collocated finite element – finite volume space- and discrete-event, asynchronous time discretization of plume spreading using the CSMP++-based ACGSS simulator (Youssef, Matthai et al., 2020, IJGGC). Left): comparison between commercial (corner-point grid, TPFA) reservoir simulator (Eclipse 100) and FECFVM and ED-FECFEM discretizations in space and DES in time (Shao, Matthai et al. 2019, JCP, 2021 COMG). Right): Schematic of DES – asynchronous evolution and parallel event processing (PEP).

By comparison with Eclipse and our previous FECFVM-based reservoir simulator, the embedded discontinuity piecewise-linear-for- saturation ED-FECFVM discretization substantially increases physical realism, overcoming the first-order vertical grid resolution dependence of Eclipse, while honoring the capillary seal related pressure- and saturation decoupling across material interfaces below which the plume spreads in the model. The interphase physics require some additional explanation:

When buoyant non-wetting CO_2 accumulates in sandstones with large pores capped by fine-grained layers with a higher entry pressure, this filtration eventually chokes off the flow of water as well. This leads to the formation of "vapour lock" = transient capillary seals, decoupling both-, pressure and saturation. Such dynamic seals can be seen in lab experiments such as the ones that Bill Benzing did at Conoco Phillips in the 90's (https://lnkd.in/gFs5KEp). We reproduce this dynamic / episodic non-steady flow accurately using a nonlinear transfer algorithm with change of variables in conjunction with the “SplitBoundary” extension of the FEM-FVM framework in CSMP++, see Nick & Matthai (2011 VZJ; 2011 TIPM), Tran et al. 2020 (AWR), Matthai & Tran (2023, AWR), Youssef et al. (2024, TIPM).

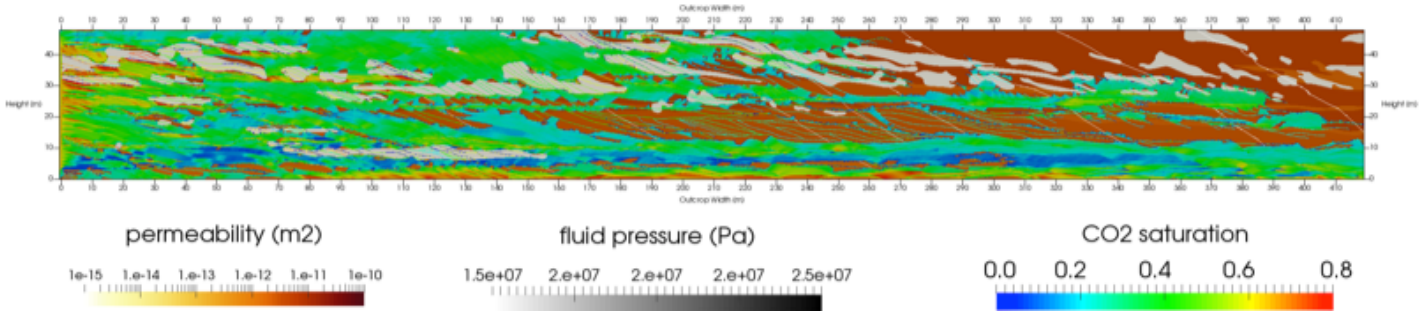


Upscaled sandbox model (2.5 x 0.8 x 0.55 km) of Chauvin et al., 2018 (AAPG Bull.), serving as a proxy of a faulted CO_2 storage site in sand-shale stratigraphy. The CO_2 is injected through a vertical well at the base of the graben structure at 1 Mt/yr; for details see Shao et al. 2020 (ECMOR-XVII). Pore-space filling histories were computed with ACGSS-CSMP++ for all layers of the stratigraphy, counting upwards from the injection layer (SL1, not shown). Permeability values refer to fault segments in shale.

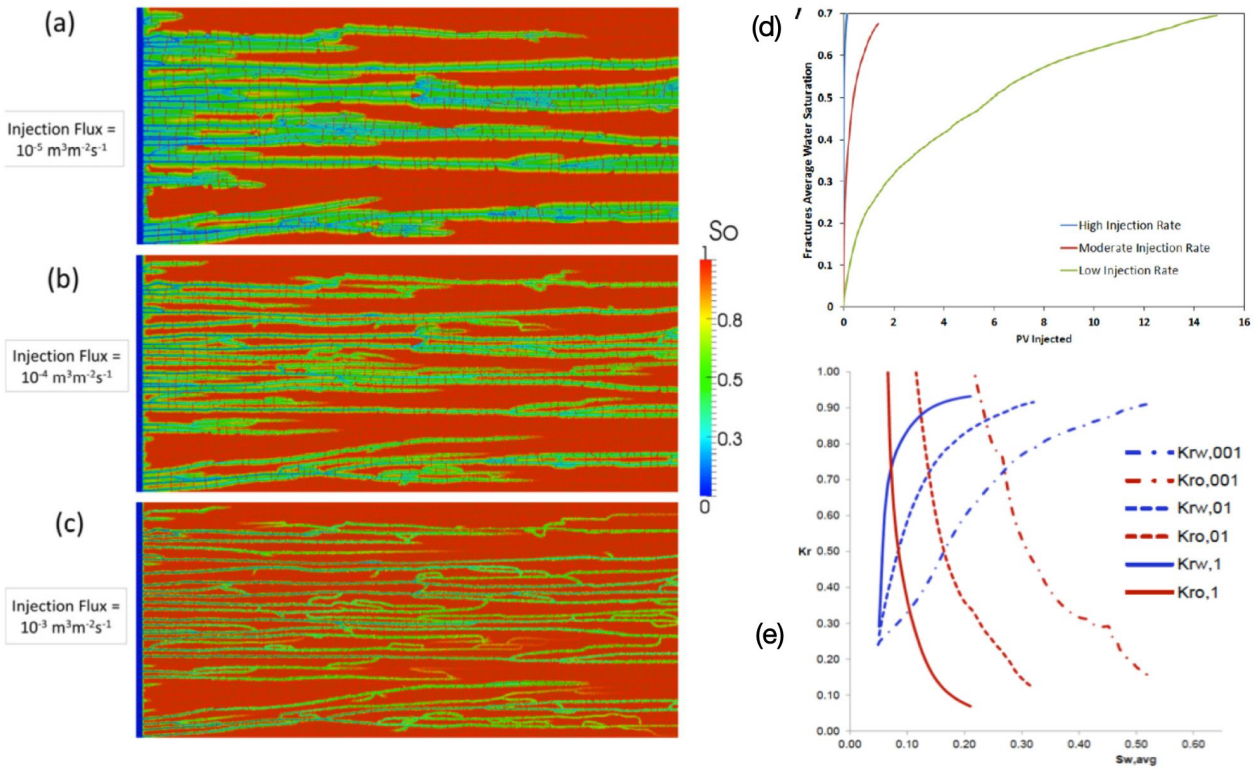
CSMP++ simulation of the breaching of a transient capillary seal caused by the accumulation of buoyant CO_2 in the footwall of a fault (cross-section 500 x 500m). When such seals are breached by over pressuring, CO_2 rapidly drains upward, while imbibition occurs in the layer below. Subsequently, this hysteretic process repeats (Tran et al., 2021, AWR). Right): simulation of the same process following CO_2 injection through a discrete vertical well completion at the base of a sedimentary graben structure (Shao et al., 2021, COMG), leading to the formation of a stacked CO_2 accumulation like that seen at Sleipner.

Since most aquifers are faulted, compartmentalization by faults and potential leakage through them are major CCS performance and safety issues on par with the risk of induced seismicity. To this day, the simulations shown here are the only realistic ones of this process, indicating that dynamic capillary seals may form across originally perfectly conductive faults, and that fault segments with cm-thicknesses and moderate permeability can still transmit CO_2 to shallower strata, resulting in stacked accumulations (similar to hydrocarbon ones), extending all the way up to the undeformed parts of the basin stratigraphy.

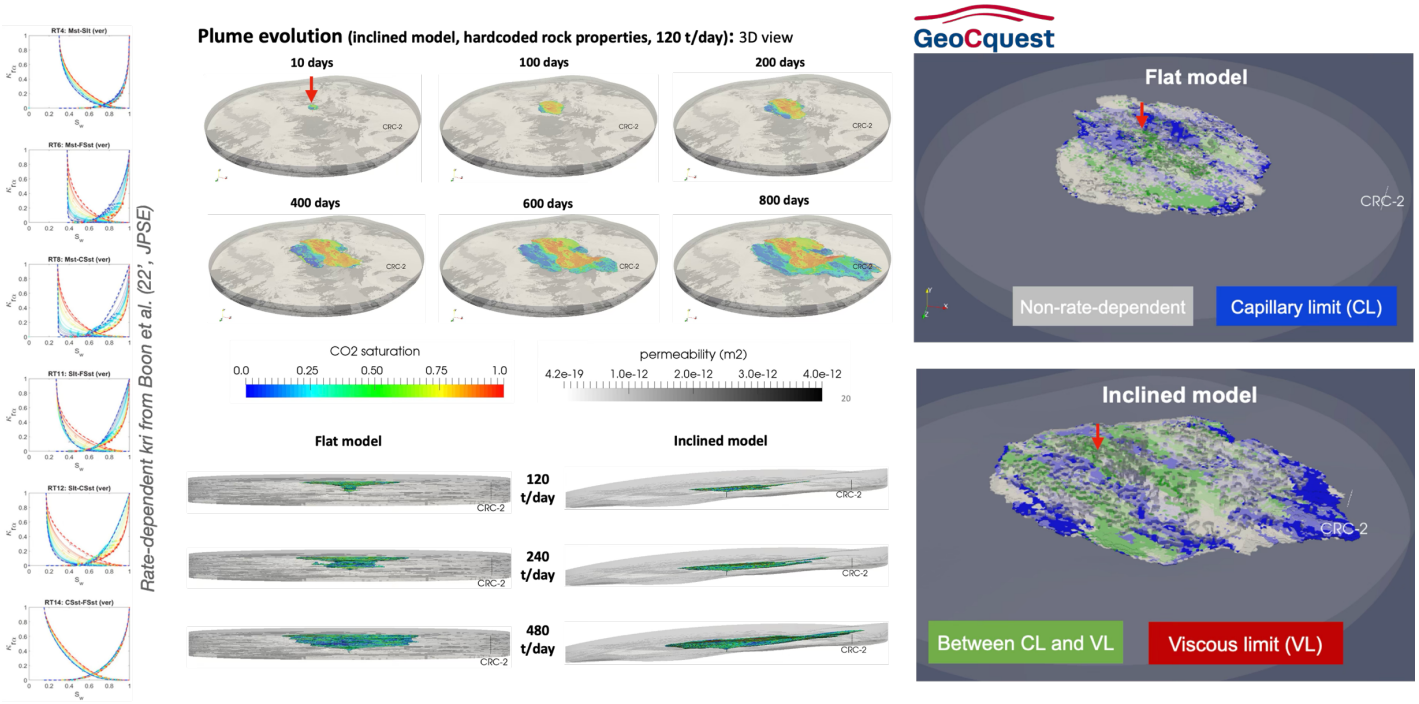
Heterogeneity and Saturation Functions. Sandstone aquifers are inhomogeneous. Bedforms, gradations, and laminations occur on all scales, giving rise to nested heterogeneity. The detailed outcrop study by Willis et al. (1999, *Sedimentology*), documents this for the Frontier FMT, WY (USA). The displayed simulation (Matthai & Burney 2018, GHGT), illustrates how high-permeability streaks at the base of this sedimentary sequence, induce fingers and lobes in the plume of supercritical CO_2 injected on the left. Caused by the interplay of reservoir heterogeneity and the high mobility ratio of $CO_{2,sc}$ and formation brine, early plume breakthrough through the right model boundary already occurs at an $S_{CO2,avg}$ of 3.5%, highlighting the poor sweep achieved as computed with this 3M cell model. This result mirrors the field observations made at Sleipner.



Which relative permeability curves should be used for highly heterogenous sandstones? - As concluded by Durlofsky (1997, SPEJ), *"...the functional form of traditional relative permeability descriptions is too limited to capture a wide variety of flow behaviors... and that an 'extended description,' in which the upscaled relative permeabilities depend on variables in addition to the phase saturations, is required."* This conclusion is well illustrated by the rate-dependent saturation functions measured in discrete fracture and matrix DFM simulations for fractured permeable rocks (e.g., Matthai et al. 2012, ECMOR), but also by relative permeability and capillary pressure curves computed for heterogeneous composite rock types by Boon, Matthai et al. (2022 IJPSE), Matthai & Tran (2023, AWR), and for cross-bedded sandstones in particular (Youssef, Matthai et al., 2024, TIPM). The latter two articles present novel embedded – discontinuity approaches with periodic boundary conditions that remove boundary artifacts such as "capillary end effects" that cannot be avoided in the laboratory experiments with heterogeneous rocks. It follows that our new numerical upscaling expands petrophysical / flow property analysis beyond what is possible with Special Core Analysis (SCAL), a cornerstone of reservoir engineering.

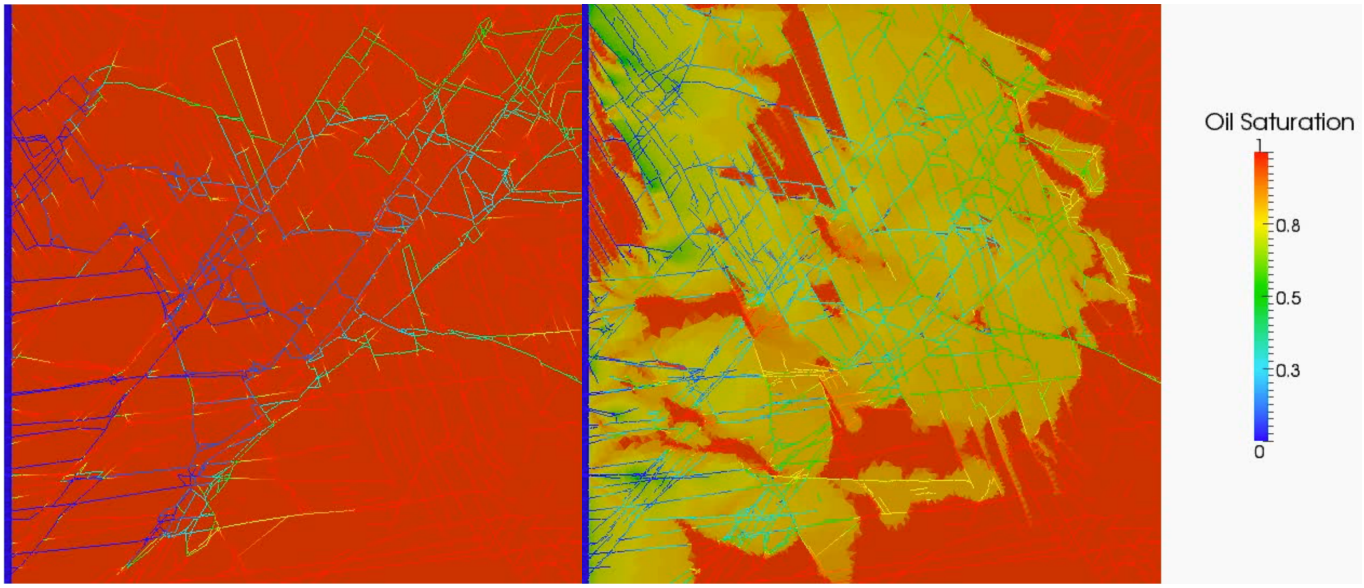


(Left): 2D water-flooding simulation, model Kilve "Bed 3" (Matthai et al. 2012, ECMOR proceedings). Oil saturation snapshots from 3 runs with different water injection rates taken once 0.5 PV were injected. a) slow rate, b) intermediate rate, and c) fast rate; d) fraction of the fracture-matrix interface area contacted by the injected water that increases with injection rate because the water enters more of the smaller fractures. e) Fracture-matrix ensemble relative permeability extracted from runs (a) to (c): solid lines mark curves for fastest injection rate (kri,1), causing rapid water breakthrough to rightsize boundary. For slow injection (kri,0.001), water breakthrough can be discerned by a distinct drop in kro at sw=0.45 (dash-dot curves).



Flow rate and flow-direction dependent saturation functions computed for 6 composite rock types from the Otway International Research Facility (OITC), Victoria, Australia, see Boon et al., (2022, IJGGC) and Shao et al. (2022, IJGGC). Left column): cross-lamination relative permeability in the capillary limit (blue) and the viscous limit (red). Center): Plume spreading around the vertical injector well CRC-3 in flat and seal-topography high-resolution (5M cell) geomodels informed by core-logging and seismic attribute tuning to determine the spatial correlation structure of porosity-permeability. The results show that the new flow-regime dependent saturation functions provoke first-order differences in the plume extent and internal saturation distributions. Right): flow regimes in the highly heterogeneous model, showing that even for the moderate injection rates of 120-480 tonnes per day, the capillary limit is exceeded in most of the permeable rock types activating the CL-VL behavior (green regions in the right figures). The gray areas flooded by the plume consist of homogeneous rock types without rate-dependent flow behaviour.

Enhanced Oil Recovery adds value to CO_2 geo-sequestration, by using the CO_2 injected into the subsurface to improve the recovery of oil. For example, CO_2 captured from the Boundary Dam Carbon power station, Saskatchewan, Canada, has been used to enhance oil recovery from the Weyburn field, a naturally fractured reservoir in the nearby Williston Basin. Supercritical CO_2 might also be a better flooding agent for mixed-wet or oil-wet naturally fractured carbonate reservoirs of the Middle East because these carbonates are wetted by CO_2 so that water flooding only displaces a small fraction of the oil stored in the rock matrix, see simulation results below, showing an 18 x 18-m top view of a mixed-wet oil-saturated fractured carbonate layer in plan view. The injection is through the sand-face section of a horizontal well on the left. While water sweeps and imbibes part of the rock matrix, oil saturations stay high. This situation could be much improved by CO_2 flooding, see Sedaghat, Matthai, and Azizmohammadi (2017, J. Pet. Sci. & Eng., and fig. 22 therein).



CSMP++ finite element – finite volume discrete fracture and matrix (DFM) simulation of water flood of a naturally fractured limestone bed through a horizontal injector well on the left (top-view, 18 x 18m window); oil saturation at 0.1 pore volumes injected (PVI). Left) oil-wet case featuring an entry pressure of the rock matrix for water, and (right): mixed-wet case.

At the 2005 SPE Reservoir Simulation symposium in Houston, we presented the first Discrete Fracture and Matrix (DFM) simulations of water-flooding with a geostatistically realistic model of a fractured reservoir (SPE93341). Multiphase flow through this model was solved using an Implicit-for pressure and saturation (IMP-IMS) method and the Algebraic Multigrid Solver for Systems (SAMGp) in a collaboration with Klaus Stueben and Tanja Clees at Fraunhofer Gesellschaft, Germany.

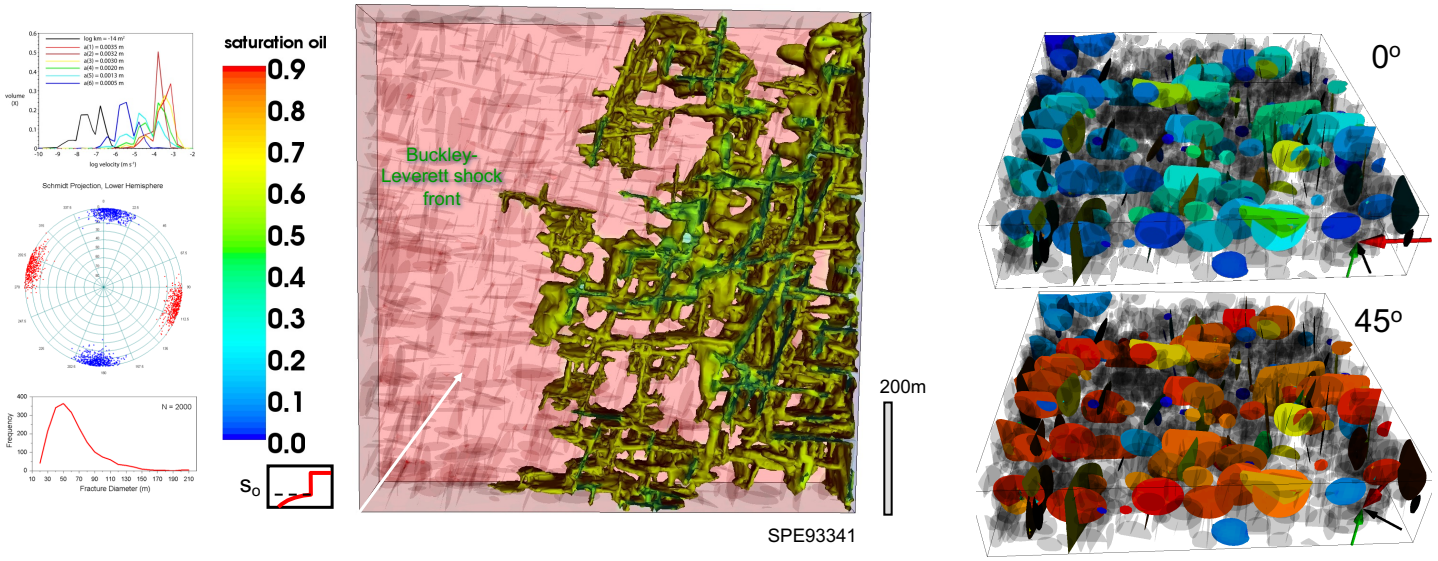
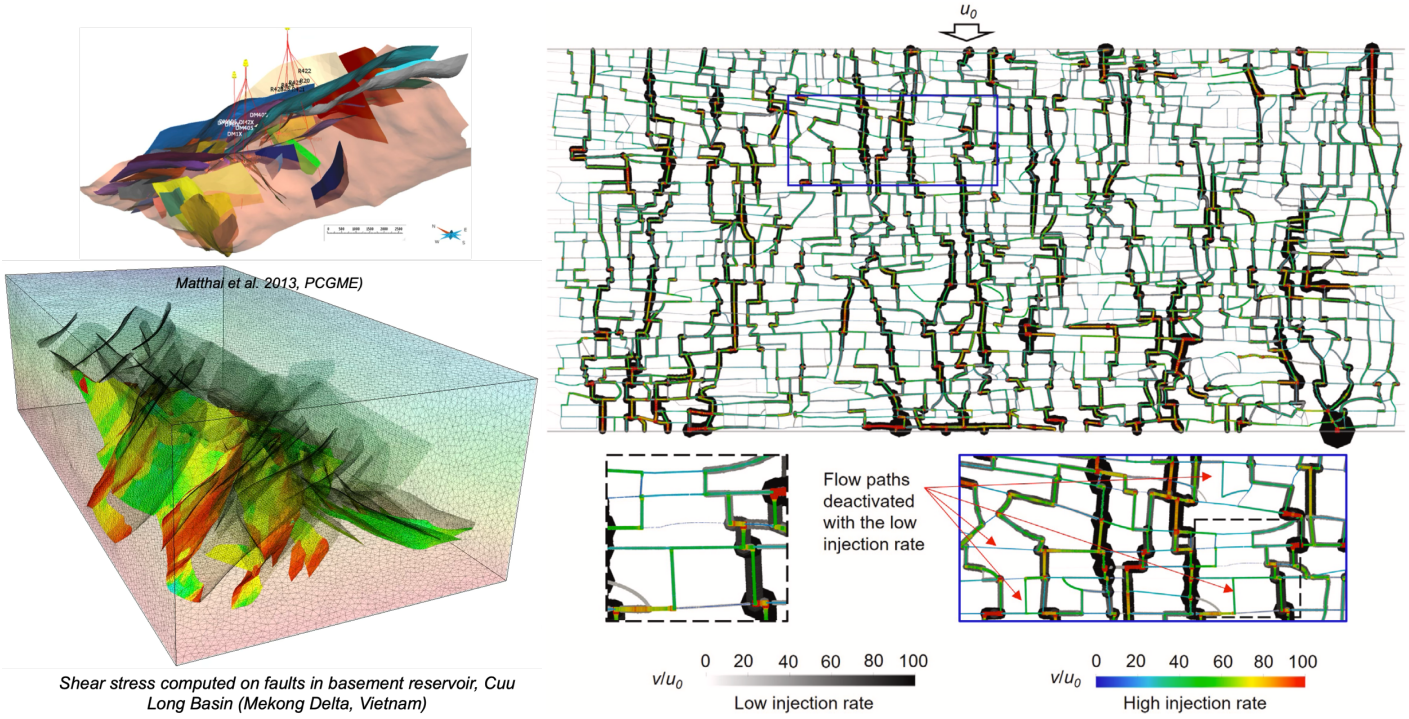


Fig. 1: Discrete fracture and matrix (DFM) flow simulation example: (left) Waterflood of 2 x 2 x 0.2 km model of a naturally fractured reservoir stochastically populated with 2000 fractures with a power-law diameter range from 5 to 180 m (Matthäi et al., 2007, SPE RE&E Journal); fracture statistics from San Andreas formation, USA, as compiled by Paul LaPointe, Golder Associates, Inc.; fracture aperture is consistent with field measurements, ranging between 0.5 and 3.5 mm and correlating linearly with fracture diameter. (right): other fracture-aperture realisations based on the far-field stress indicating the impact of stress tensor rotations (red-green-black vectors for $S_{H,max}$, $S_{intermediate}$, and $S_{h,min}$, respectively) on fracture aperture.

Since the conclusion of the itf-JIP on naturally fractured reservoirs in 2013, I have conducted industrial research and reservoir engineering field-studies for the Vietnamese oil industry and Vietnamese Petroleum Institute with a focus on the faulted basement reservoirs in the Mekong Delta. Together with the company NFR Studies Gmbh we built, and history matched several newly constructed fully unstructured reservoir models in terms of pressure, water cut, water salinity, and production temperature using our original CSMP++ based reservoir simulator (see example model below). This work also necessitated modelling the *in situ* stress and its impact on fault permeability.

This work provided the first incentive to move on from the laminar-creeping Stokes formulation for multiphase fracture flow to a Reynolds' averaged NS model considering time-averaged inertia effects (Matthai et al., 2024, AWR), and Detached Eddy Simulation to capture vortex shedding and fluid rheology effects within fracture networks (Bui & Matthai, 2024, ECMOR). For the latter, we validated initial high-resolution single-phase flow simulation results with physical experiments, opening an exciting new field in HPC and applied research and development of high relevance for ARAMCO and subsurface engineering applications in Saudi Arabia. Importantly, these fracture-network models behave fundamentally different from what had previously been predicted by extrapolating findings from single fracture research. Among several fundamentally new and important findings, this research demonstrates that inertia effects manifest already at flow velocities of a few cm per second.



Shear stress computed on faults in basement reservoir, Cuu Long Basin (Mekong Delta, Vietnam)

Examples of recent research on fractured / faulted reservoirs. Left): shear stress acting on a complex array of geologic faults in a THMC model of a fractured basement reservoir off-shore Vietnam as computed from the *in situ* stress using direct measurements in well and GPS stations as constraints. Right): Flow-rate dependent flow paths through a 15M cell Reynolds-averaged NS model of a fracture network mapped in outcrop (Matthai et al., 2024, AWR).